

NSDF-Data student guide 2: your first analysis

2026-10-05

This guide takes you from a finished setup (guide 1) to your first real pulse analysis: download one dataset, load it, and reproduce a number from *Note 2*. Expect about two hours. The data is about 34 MB, so the download is quick.

Work on your personal branch (`student-<yourname>`). Open a terminal and activate the environment: `conda activate darkmatter_cli_env`.

1. Read the notes first (30 minutes)

Read these, in this order, on the notes site:

1. **Note 1**, *Run76 series catalog*. What is a series, a dump, a trace?
2. **Note 1a**, *python/ library structure*. What each module does.
3. **Note 2**, *Noise-pulse selection*. What `bstd` and `ratio` are, and how the 179-trace noise selection is made.

You do not need to follow every number in Note 2 yet. Learn the words: *series*, *dump*, *trace*, *channel*, *detector*, *pretrigger window*, *baseline*.

2. Download one dump

The dataset we use throughout is series 07221203_2025, dump F0001:

```
nsdf-cli download 07221203_2025_F0001
```

Downloaded files go into an `idx` folder in your home folder, as `idx/07221203_2025_F0001`. Do not move them, and do not commit them to Git. The download takes a few minutes. If it fails, check your network and run the same command again. If the folder already exists the tool skips the download, so delete the empty folder first if it is empty.

3. Load the data and count the traces

Open a Python session in the repository (or a new notebook in `R76/analysis_notes/`) and run:

```
import sys
from pathlib import Path
sys.path.insert(0, str(Path.cwd().parents[1] / "python")) # from R76/analysis_notes/

from nsdf_dark_matter.idx import load_all_data
import pulse_io as pio
```

```
cdms = load_all_data(str(Path.home() / "idx" / "07221203_2025_F0001"))
ids, pulses = pio.load_channel_batch(
    cdms, pio.filter_by_detector(cdms.get_detector_ids(), 0), 0)
print(pulses.shape)
```

You should see (1517, 4096): 1517 traces of 4096 samples on Detector 0, Channel 0. Each sample is 1.6 microseconds. If your count is different, stop and check the dump name and the detector number.

4. Look at a trace

```
import matplotlib.pyplot as plt
plt.plot(pulses[0])
plt.xlabel("sample (1.6 μs each)"); plt.ylabel("ADC counts")
plt.show()
```

Most traces are flat noise. Look at a few: `pulses[1]`, `pulses[2]`, and so on.

5. Reproduce one number from Note 2

Note 2 says that for this dump the noise selection leaves **179 traces**. The steps are in the Note 2 notebook. Start from the library instead:

```
from dataclasses import replace
from pulse_config import DEFAULT_CONFIG
import pulse_quantities as pq
import pulse_cuts as pc
import numpy as np

c500 = replace(DEFAULT_CONFIG, pretrigger_samples=500)
sel = pc.excursion_band(pulses, c500) & (np.log10(pq.bstd(pulses)) < 0.5)
print(sel.sum())
```

You should get **179**. If you get something else, that is worth knowing, so tell the project lead the number you got and the steps you used. Do not change the library to make the number match.

6. Write it down

Put your work in a notebook named with your initials and today's date, for example `student-<initials>-2026-10-05.ipynb`, in `R76/analysis_notes/`. Keep the cells short and label each one with a sentence saying what it does. Commit it to your branch:

```
git add R76/analysis_notes/student-<initials>-2026-10-05.ipynb
git commit -m "First analysis: load dump 1 and reproduce the 179-trace selection"
git push -u origin student-<yourname>
```

Done when

- You downloaded 07221203_2025_F0001 and loaded (1517, 4096) traces.
- You plotted at least three traces and can say what the flat ones look like.
- You reproduced 179, or wrote down what you got and why you think it differs.

- Your notebook is committed to your branch.

Common problems

- **ModuleNotFoundError: nsdf_dark_matter:** you are not in the `darkmatter_cli_env` environment. Run `conda activate darkmatter_cli_env` and restart the kernel.
- **ModuleNotFoundError: pulse_io:** the `sys.path` line points at the wrong folder. Print `Path.cwd()` and adjust it.
- **Plots show a huge spike at the start of a trace:** that is a leading glitch in the first few samples. It is real, not a bug. Note 2a explains it.
- **Do not run `07221203_2025_dump1_noise.ipynb` unless the project lead says it is fine.** It overwrites a shared archive file.

Session Info

Guide version: 2, 5 October 2026. Questions go to the project lead.

Key links

- Note 1: <https://villano-lab.github.io/NSDF-Data/notes/note-01-run76-series-catalog.html>
- Note 1a: <https://villano-lab.github.io/NSDF-Data/notes/note-01a-pulse-library-structure.html>
- Note 2: <https://villano-lab.github.io/NSDF-Data/notes/note-02-noise-selection.html>
- Repository: <https://github.com/villano-lab/NSDF-Data>
- Student guide 1 (setup): <https://villano-lab.github.io/NSDF-Data/students/student-guide-1-setup.pdf>